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Track 01

AI Systems & Workflow Innovation Challenge

Team Name : AutoVault Intelligence

Problem Statement: Professionals suffer from application fatigue due to constant switching between multiple email platforms. Meanwhile, centralized cloud providers compromise privacy by storing decryption keys on remote servers, and standard spam filters fail to block invasive tracking pixels and sophisticated phishing scams.

Team Members : Vatsal Agarwal

Problem Definition

- What problem are you solving?
 - **Data Exploitation & Privacy Gaps:** Centralized cloud platforms inspect private message contents on remote databases, removing true user data sovereignty.
 - **Invasive Tracking:** Senders hide invisible 1x1 pixel tags in HTML mail text to automatically harvest user locations, open timestamps, and device signature profiles.
 - **Post-Delivery Web Scams:** Attackers execute "Time-of-Click" redirect mutations, sending a benign URL that transforms into a phishing site after bypassing cloud scanners.
 - **Rigid Keyword Search Failures:** Standard inbox indexers fail to surface files or text unless users remember literal keyword matching or precise strings.
- Who experiences this problem?
 - Corporate professionals, technical builders, developers, and power users managing high-volume communication feeds across multiple accounts.
- Why is the current approach insufficient?
 - Standard image proxies merely mask the host IP address but still let the tracking pixel load from the sender's server, confirming active user presence.
 - Cloud-side link scanning acts only at the immediate moment of delivery, leaving users entirely vulnerable when they physically click the mutated link later.

Opportunity &

Vision

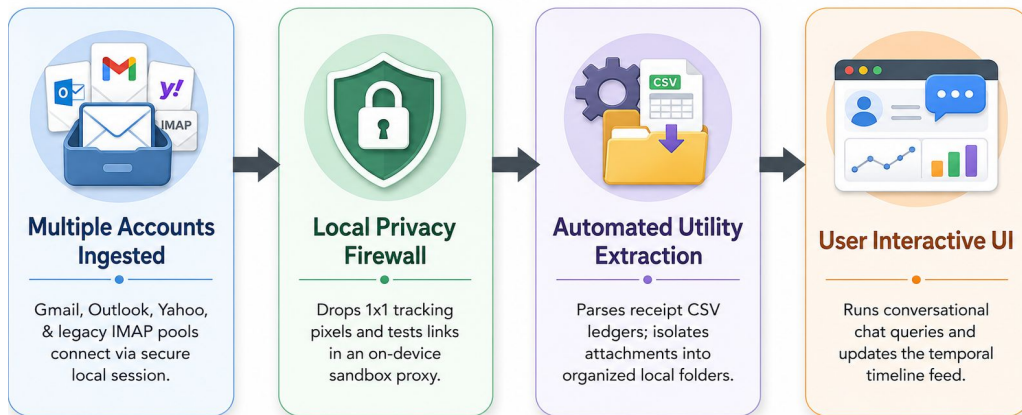
- Why is this an important opportunity?
 - There is a massive, unaddressed market gap for an email terminal that delivers advanced, automated AI utilities without forcing users to sacrifice privacy.
 - Moving the data aggregation, vector math, and security screening entirely to client-side hardware creates immediate user trust and total data isolation.
- What future state are you enabling?
 - A zero-knowledge workspace where multiple legacylike and modern email accounts are unified into a fast, single-timeline interface on the machine.
 - An intelligence engine that turns passive mail into automated actions parsing expenses into local Excel files and cataloging document sheets natively.

Solution Overview

- What is your proposed solution?
 - Kinetic Mail: A client-side, zero-trust email client that combines multi-mailbox aggregation, localized privacy firewalls, and natural language AI.
- What makes it AI-native rather than AI-assisted?
 - The core system architecture is built around an on-device text-embedding vector index and an optimized local light-LLM pipeline.
 - AI acts as a core system component rather than a generic text plug-in—automatically executing contextual intent tab routing and deep conversational chatbot queries.
- Which existing Redrob capability, workflow, or user journey does your solution build upon?
 - It directly extends Redrob's AI-native ecosystem spanning productivity, communication, and workflow automation by establishing a highly secure, data-private terminal for day-to-day business messaging.

User Journey / Workflow Diagram

- **How does a user interact with the solution?**
 - The user signs into their Gmail, Outlook, Yahoo, or legacy accounts once; refresh tokens are saved strictly on their device hardware.
 - The client silently strips tracking pixels and handles security sandbox checks in the background as new mail arrives.
 - The user opens the app to see a clean, unified dashboard featuring smart tabs, a financial ledger list, and an organized document folder.
- **How does information flow through the process?**
 - Incoming Email Stream → Local HTML Pixel Stripper → Background Sandbox Proxy (Pre-screens URLs/QRs) → On-Device Context Parsing Engine → UI Presentation Tabs.
- **Where does the solution integrate with Redrob experiences?**
 - It pipes clean, parsed communication text and scheduling strings directly into Redrob's task management and recruitment workflow modules.



AI Logic & Decision Flow

- Where does AI intervene?

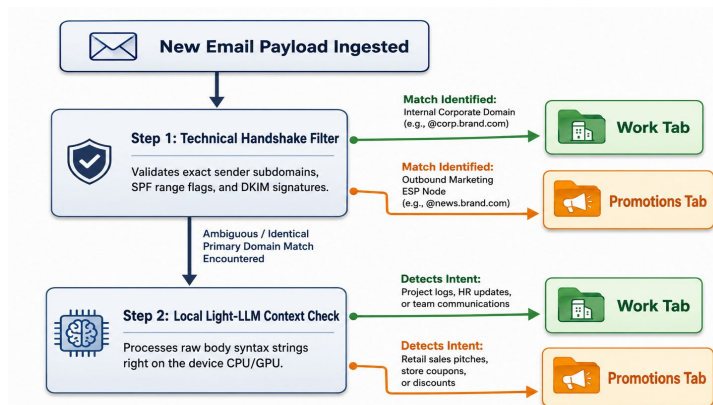
AI intervenes during the initial incoming message parsing stage to determine semantic intent and solve classification confusion.

- How are decisions made?

Decisions are made sequentially by combining deterministic cryptographic header checks with an on-device fallback light-LLM.

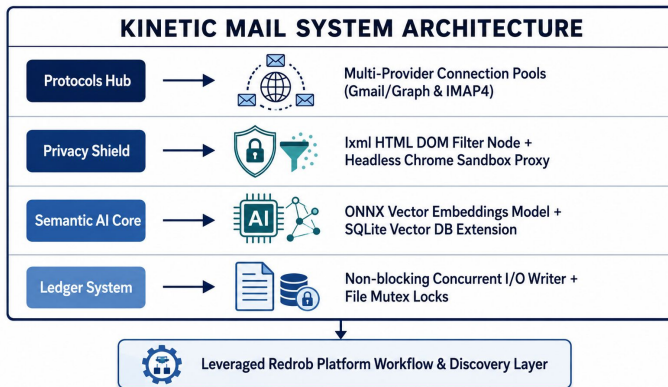
- How do agents/models/reasoning systems interact?

The model acts as a secondary verification step: if the domain structure is ambiguous, the local light-LLM reads the raw body text syntax to differentiate work items from promotions.



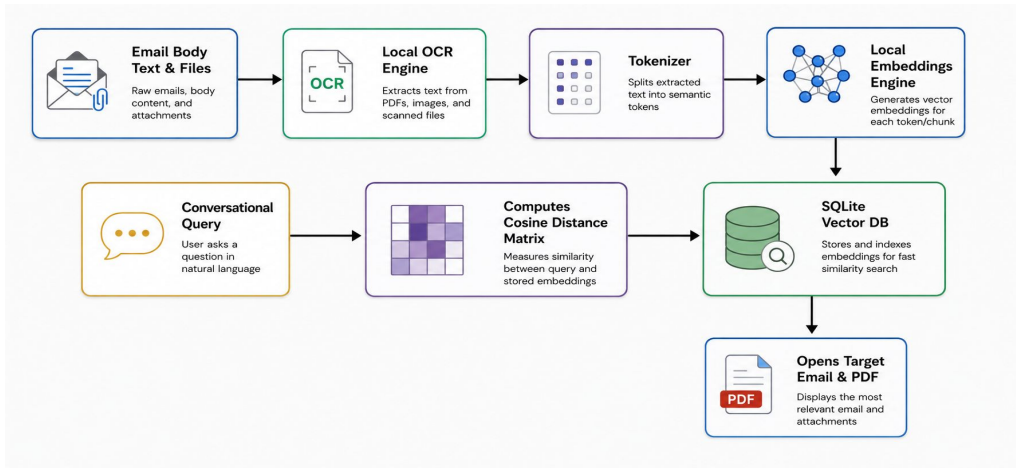
System Architecture

- What components make up the system?
An Ingestion Hub, an Ixml HTML DOM Filter Node, a Headless Sandbox Proxy, an ONNX Embedding Engine, and a thread-safe Concurrent File I/O Writer.
- How do services interact?
Ingested data is streamed through memory-mapped parsing layers, where security modules clean the text before indexing or ledger writing takes place.
- Which Redrob systems, workflows, or services are leveraged?
It pairs with Redrob's native workflow recommendation layer, using secure message data to trigger automated project management actions.



Data, Context & Intelligence Layer

- What data powers the solution?
Raw unstructured email text strings, email metadata headers, and text extracted from attachments using local OCR tools.
- How is context retrieved, stored, or utilized?
Context is tokenized and stored on-device as mathematical vectors inside an embedded database instance, enabling natural language search queries.
- How does existing Redrob context improve the experience?
It pulls historical task context from Redrob to better understand user search queries, increasing chatbot accuracy without using tracking cookies.



Scalability & Technical Feasibility

- How would this be implemented?

As a fast cross-platform desktop app utilizing an Electron framework tied tightly to optimized local Python-compiled execution modules.

- How does the system scale?

It balances scaling perfectly. Because all vector math, security filtering, and file indexing run locally on the user's physical device hardware, our central databases face zero operational processing bottlenecks as the platform grows.

- What technical challenges exist?

Local AI model calculations can cause minor CPU latency during high-volume setups. We fix this by running all text embedding tasks asynchronously in separate, non-blocking background threads.

Redrob Ecosystem Integration

- **Which existing Redrob capabilities are being leveraged?**
Redrob's core pipeline for workplace task management, network discovery, and recruitment communication channel mapping.
- **What new capability does your solution introduce?**
A unified data ingestion engine, a biometric hardware-isolated attachments vault, and an automated multi-CSV expense accountant tool.
- **How does it strengthen the overall Redrob ecosystem?**
It introduces an enterprise-grade B2B Domain Verification Registry where brands pay a fee to secure an un-spoofable Brand Verification Badge, boosting open rates while completely undercutting phishing attempts.
- **What additional opportunities become possible after implementation?**
Redrob can leverage this secure terminal to build automated autonomous agents that securely schedule meetings and screen candidates directly via local email analysis.

Impact & Success Metrics

- What measurable outcomes are expected?
 - 100% deletion of hidden tracking pixel elements before an email renders visually.
 - Zero-latency execution of safe, sandboxed redirect checks at user click-time.
 - Near-instant retrieval of buried files and receipts using plain conversational text prompts.
- How will success be tracked?

Tracked via on-device logging tables calculating pixel deletion volume, links caught by the sandbox proxy, and query similarity match metrics.
- What value is created for users and Redrob?
 - For Users: Completely automated personal bookkeeping sheets, total information privacy, and an uncluttered workspace.
 - For Redrob: Drives immense platform adoption from developers, enterprise clients, and data-sensitive industries, creating strong ecosystem lock-in.

Future Roadmap

- How could this evolve over 2–3 years?
 - Year 1: Roll out the core ingestion client across Gmail/Outlook along with the automated thread-safe multi-CSV financial writer.
 - Year 2: Integrate the local embedding chatbot interface alongside on-device OCR file vault categorization.
 - Year 3: Fully launch the B2B verification badge registry subscription models for mid-market corporate brands.
- What future capabilities can be unlocked?

Context-aware, automated local reply drafting that safely learns the user's historical writing habits across all accounts without cloud exposure.
- What broader vision does this support?

Building a highly reliable digital system that handles everyday communications securely and runs on completely sovereign client-side infrastructure.



KINETIC MAIL

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THANK YOU